---

title: “Law as Field Constraint: Regulation, Rights, and the Topology of Social Attractor Space” subtitle: “Fractal Programme — Regulatory Scale (12–14)” author: “Alistair Johnson” orcid: “0009-0007-2194-0850” institute: “Independent Researcher, Zurich, Switzerland” date: “2026” lang: en-GB bibliography: ../../paper/bibliography.bib csl: ../../paper/apa-7th.csl abstract: | We apply the Universal Somatic Field (USF) framework to the regulatory scale (scales 12–14, 10 –10 m) to derive a formal model of legal and regulatory systems. The central identification: a legal norm is a constraint on the attractor space of the social field — it eliminates certain attractor basins (prohibited states) and deepens others (protected states). Rights are formally defined as topological invariants of the legal field: states that are protected against perturbation by positive winding number around a regulatory constraint surface. Regulation is the volitional source term $J_{\text{law}}(t)$ applied at the social field level — an external driving force that reshapes the energy landscape accessible to social actors. We derive three structural theorems: (1) the rule of law is equivalent to the ergodicity condition on the social field — every state accessible under the law is reachable from every other; (2) legal uncertainty produces attractor fragmentation — multiple incompatible social equilibria with no shared basin boundary; (3) the difference between a law and a norm is the difference between a hard constraint (removing attractor basins entirely) and a soft constraint (raising the energy of disfavoured states). These identifications connect jurisprudence to physics at the structural level, with testable consequences for institutional design.